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Weather

WEATHER SUPPORT

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This publication implements Department of the Air Force Policy Directive (DAFPD) 15-1, *Weather Operations*; Air Force Manual (AFMAN) 15-111 (IC 1 26, Jan 24), *Surface Weather Observations*; AFMAN 15-124, *Meteorological Codes*; DAFMAN 15-129, *Air and Space Weather Operations*; and Federal Meteorological Handbook 1 (FCM-H1), *Surface Weather Observations and Reports.* Refer recommended changes and questions about this publication to the OPR listed above using the Department of the Air Force (DAF) Form 847, *Recommendation for Change of Publication;* route DAF Forms 847 from the field through the appropriate chain of command. The authorities to waive wing/unit level requirements in this publication are identified with a Tier (T-0, T-1, T-2, T-3) number following the compliance statement. See Department of the Air Force Instruction (DAFI) 90-160, *Publications and Forms Management,* for a description of the authorities associate with the Tier numbers. Submit requests for waivers through the chain of command to the appropriate Tier waiver approval authority, or alternately to the Publication OPR for non-tiered compliance items. Ensure that all records created because of processes prescribed in this publication are maintained in accordance with Air Force Instruction (AFI) 33-322, *Records Management and Information Governance Program*, and disposed of in accordance with Air Force Records Information Management System (AFRIMS) Records Disposition Schedule (RDS). The use of the name or mark of any specific manufacturer, commercial product, commodity, or service in this publication does not imply endorsement by the Air Force.

This instruction, in conjunction with Air Force Weather publications and appropriate supplements, establishes requirements and responsibilities for organizations providing weather support to 52d

Fighter Wing (52 FW) organizations. It provides policy and procedural guidance for the functions of weather support provided by the 52d Operations Support Squadron, Weather Flight (52 OSS/OSW), and establishes responsibilities for other 52 FW organizations. It provides general information for weather services, including weather observations and forecasts, watches, warnings, and advisories, space weather supported services, dissemination of information, and reciprocal support. It provides guidance for in-garrison weather support and weather support at deployed locations. It applies to all personnel assigned or attached to the 52 FW and serves as a guide for its supported/supporting units, as well as outlines how the 52 OSS/OSW coordinates with the 557th Weather Wing (557 WW), Operational Weather Squadrons (OWS).

SUMMARY OF CHANGES

This document has been substantially revised and must be completely reviewed. Major changes include AFI 15-114 being rescinded as of June 2021, AFI 15-127 being rescinded as of 2024, and AFMANs 15-129v1 and 15-129v2 consolidated rewrite as DAFMAN 15-129 current as of September 2023. Many changes only effect internal weather operations, specifically the relationship between OWS and OSS Weather Flights. (DAFMAN 15-129). Changes include expansion of overall roles and responsibilities for HAF, MAJCOMs and weather units, update to the Special Assistance Requests (SAR) procedures to match current 557 WWG processes, a reduction in tier ratings to allow commanders more flexibility, and clarification on hazard charts, airfield support, RPA procedures, FWBs, and ANG support to address common issues across the weather career field. This interim change revises AFMAN 15-111 by (1) updating Special (SPECI) criteria for snow, (2) including temperature readings on discontinuity sensors, (3) clarifying rounding of figures and values, (4) Runway Visual Range (RVR) active runway clarification, and (5) deleting “eyes forward” OWS duty. A margin bar (|) indicates newly revised material.

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Chapter 1

GENERAL INFORMATION

1.1. Purpose. The 52 OSS/OSW tailors and integrates weather information in support of 52 FW operations. 52 OSS/OSW may provide and/or assist in arranging weather support services for tenant units within 52 FW areas of responsibility.

1.2. Implementation. This instruction establishes the responsibilities and procedures for areas of weather support that must be coordinated at the local level to meet mission needs and requirements. This instruction is followed during peacetime and wartime operations, unless superseded by Emergency War Orders. Coordination on this instruction constitutes acceptance in lieu of a Memorandum or Letter of Agreement (MOA/LOA) unless one is otherwise required.

1.3. Concept of Operations. 52 OSS/OSW works in partnership with the 557 WW and supporting OWS, IAW DAFMAN 15-129, to tailor operational and strategic level weather products for the operational user and integrate decision-grade information on weather and environmental impacts to military operations. Full aspects of support are documented in the Installation Data Plan (IDP) between the supporting OWS and 52 OSS/OSW and are reviewed annually at a minimum. Changes to this instruction affecting the supporting OWS provided services will be coordinated between the two agencies.

1.4. Duty Priorities. The duty priorities for the 52 OSS/OSW and the supporting OWS are documented in the IDP. The Duty Priorities table exists to match and balance manning and mission critical tasks during peak work periods prone to task saturation. These duty priorities were developed IAW DAFMAN 15-129. Weather personnel will use good judgment in complying with these duty priorities, especially where there is imminent danger to equipment, life and/or property.

1.5. Equipment.

1.5.1. Primary Systems.

1.5.1.1. AN/FMQ-19. The AN/FMQ-19 is the fixed based weather observing system (FBWOS) for Spangdahlem AB. It is an integrated system of multiple weather sensors and data automation components across the airfield that continually measure environmental conditions. These sensors and components provide responsive, reliable, accurate, real-time weather information in support of flight operations. The FBWOS samples, measures, and reports: temperature, wind speed, wind direction, visibility, runway visual range (RVR), cloud base heights and sky coverage, surface pressure, liquid equivalent precipitation accumulation, and ice accretion during freezing precipitation. **Table 1.1** lists the FBWOS sensors and locations in association with the Spangdahlem AB runway. The Terminal Data Acquisition Unit (TDAU) and Moxa Ethernet switch are in Building 195.

Table 1.1. FBWOS Sensors and Locations.

Primary Sensors (22 end of runway)	Secondary Sensors (04 end of runway)	Mid-Field Sensor (Mid-field)	Valley Sensor (22 end, off-base)

Wind Speed Lightning Rain Gauge Relative Temperature/Humidity Freezing Rain Barometers Ambient Light Visibility Precipitation-Identification Ceilometer	Wind Speed Ceilometer Visibility	Visibility	Ceilometer

1.5.1.2. Joint Environmental Toolkit (JET) & Sensor Collection Appliance (SCA). JET and SCA are the primary communication systems used by USAF weather units. They use centralized servers and desktop computers to transmit data to and receive data from weather agencies worldwide via the internet and dedicated communication links.

1.5.1.3. Air Force Weather Web Service (AFW-WEBS). AFW-WEBS is an online portal maintained by the 557 WW that provides access to various weather products and tools (e.g., model data, satellite data, alphanumeric products, etc.). AFW-WEBS can be used as a back-up dissemination system in the event of a SCA outage. Access is dependent on LAN availability.

1.5.1.4. Air Force Weather Bridging Environmental Intelligence for Responsive Operational Support (BIFRÖST) Portal. The AFW-BIFRÖST Portal is a web-based source for trusted environmental data, products, services, and tools. Over the next year, the 557 WW will slowly migrate capabilities from JET, AFW-WEBS, and OWS web pages to the BIFRÖST Portal.

1.5.1.5. Mark IVB. Mark IVB provides global, real-time meteorological satellite imagery that can be interrogated, enhanced, and customized. It provides range, MOA, and airfield overlays and greatly increases situational awareness. Mark IVB is dependent on LAN availability.

1.5.1.6. Weather RADAR. Unlike in the continental US, Spangdahlem AB has no fix-based weather RADAR capability. 52 OSS/OSW relies on host nation radar infrastructure viewable via public domains and AFW-WEBS composite radar products. The nearest radar to Spangdahlem AB is operated by the Deutscher Wetterdienst near the village of Neuheilenbach, approximately 18 KM north-northwest of the Spangdahlem.

1.5.2. Deployable and Back-up systems.

1.5.2.1. Portable Doppler Radar (PDR). 52 OSS has coordinated temporary installation of a PDR which was sourced from the 7th Combat Weather Squadron. The PDR is currently in testing and evaluation as a supplement to commercially available radar composites.

1.5.2.2. TMQ-53. The TMQ-53 is a tactical/deployable weather system with capabilities like the FMQ-19. It can be used as a back-up system in the event of an FBWOS outage. UTC-postured (deployable) TMQ-53s are only used as back-up permanent observing

systems at controlled airfields and will not be used as, or replace, the permanent airfield observing system.

1.5.2.3. Advanced Micro Weather Sensor (AMWS). The AMWS (AMWS) is a remote, unattended, fully operational weather sensor capable of rapid deployment. It lacks the level of calibration and accuracy of the TMQ-53 but is much more compact and lightweight. It is ideal for short-notice and small footprint deployments as a back-up to pre-existing sensor infrastructure, or as a short-term sensor in austere environments or ACE locations.

1.5.2.4. Laser Range Finder. Laser range finders are used to build visibility charts for in-garrison and deployed operations. It can also be used as a back-up method to estimate cloud base heights.

1.5.2.5. Kestrel 4500/5500. The Kestrel is a hand-held weather sensor that can be used to measure various weather parameters in the event of an FBWOS outage.

1.6. Limitations.

1.6.1. Limitations and Challenges. 52 OSS/OSW limitations and challenges for forecasting at Spangdahlem AB are described in the Forecast Reference Material and the IDP. A copy of the latest documents may be requested from the 52 OSS/OSW.

1.7. Release of Weather Information to Non-DoD Agencies and Individuals. 52 OSS/OSW does not provide weather support or information to non-DoD organizations or to the general public except as authorized by DoD Directive Joint Ethics Regulation (5500.7-R) or the 52 FW/CC (or designated representative). In addition, 52 OSS/OSW personnel will not release weather data to outside agencies unless coordinated through 52 FW Public Affairs (PA) and 52 FW Staff Judge Advocate (JA) legal offices, 52 OG/CC, and the 52 OSS/CC.

1.8. Contact Information. 52 OSS/OSW can be reached during and after airfield hours at DSN: 314-452-6749 (Comm: +49-6565-61-6749), or by email (**52oss.wx@us.af.mil**). You may also visit our SharePoint (**https://usaf.dps.mil/sites/52og/OSS/OSW/SitePages/Home.aspx**) for basic weather information and feedback forms.

Chapter 2

STAFF INTEGRATION FUNCTION

2.1. Roles and Responsibilities.

2.1.1. The 52 OSS/OSW Flight Commander (Flt/CC) ensures overall operational capability of the flight by providing intent, direction, and resources in alignment with 52 FW priorities. The Flt/CC is responsible for the conduct and direction of the 52 OSS/OSW.

2.1.2. The 52 OSS/OSW Wing Weather Officer (WWO) exploits and makes weather a force multiplier by providing liaison and briefing services to all 52 FW agencies.

2.1.3. The 52 OSS/OSW Flight Chief provides technical leadership and oversees daily weather operations in support of base and flying operations. The Flight Chief adapts resources to mission requirements and manages weather services and equipment. Lastly, this individual is responsible for all training and readiness activities and the technical health of the flight.

2.2. Standby Forecaster. Outside of controlled airfield hours, the 52 OSS/OSW continues to provide resource protection services for Spangdahlem AB. A standby forecaster can be recalled to their post under circumstances outlined in Standard Operating Procedure (SOP) AWS 09-Standby Procedures.

2.2.1. The standby forecaster will respond to all Spangdahlem AB weather related calls received when on standby status as required. The following circumstances will be used as general guidance for reasons to recall the standby forecaster outside of controlled airfield hours:

2.2.1.1. Support is required for an exercise, alert, in-flight emergency, or any other contingency.

2.2.1.2. When a toxic corridor is required for HAZMAT response. The standby forecaster will remain on-duty until notified by the 52 Command Post (52 CP) or 52 Civil Engineering Squadron Emergency Management (52 CES/CEX) that the threat is over.

2.2.1.3. Any operation determined by the 52 OG/CC to require weather support.

2.2.1.4. When criteria in **Table 2.1** have occurred or are forecasted to occur.

2.2.2. When notified, standby forecasters should make every effort to arrive within 1 hour of notification, or as soon as safely possible to assess local conditions or to augment the FBWOS IAW AFMAN 15-111 and local procedures.

2.3. Severe Weather Action Plan (SWAP). The SWAP ensures sufficient personnel are available during potential or actual severe weather events. 52 OSS/OSW is responsible for conducting annual exercises to test SWAP capabilities.

2.3.1. The 52 OSS/OSW member on shift or standby will contact the Severe Weather Action Team (SWAT) member to alert them to the possibility of needing assistance when observing the first indication of the following criteria:

2.3.1.1. A weather watch or warning for any criteria in the SWAT column **Table 2.1** is issued, if the criteria is expected to occur, is actively occurring.

2.3.1.2. Upstream station(s) within 100 NM are reporting severe weather.

2.3.2. If the SWAT forecaster determines it is necessary, SWAP will be implemented, and SWAT personnel will immediately report to the weather station. A 52 OSS/OSW recall will be directed for more personnel as the situation warrants.

2.3.3. See **Chapter 5**, Weather Watches, Warnings, and Advisories for more information about weather warnings.

Table 2.1. SWAT Matrix.

CRITERIA	SWAT
Tornadoes	X
Severe Thunderstorms Hail ≥ ¾ inch & Winds >50kts (including gusts)	X
Non-convective Winds ≥ 50kts (including gusts)	X

2.4. Operational Reports. IAW AFMAN 10-206 and AFMAN 10-206 USAFE-AFAFRICASUP, *Operational Reporting*, 52 OSS/OSW will provide the 52 CP with weather information needed to report impacts of weather to higher headquarters.

2.4.1. 52 OSS/OSW will contact the 52 CP immediately if any of the criteria in **Table 2.2** are met:

Table 2.2. Severe Weather Reporting Criteria.

Tornado
Severe Thunderstorms (hail ≥ ¾ inch & winds ≥ 50 knots (to include gusts)
Non-convective winds ≥ 50 knots (to include gusts)
Any damage or injury caused by weather phenomena

2.4.2. At the conclusion of the severe weather event, 52 OSS/OSW will review and summarize the severe weather, then forward information to the 52 OSS/CC and 52 CP within 24 hours of the occurrence. 52 OSS/OSW leadership will provide or arrange delivery of hard copies of relevant forecasts, observations, watches, warnings, and equipment status to the 52 CP to be included in an OPREP-3. If an OPREP-3 is generated, the information will also be provided to USAFE A3/A3CW.

2.5. Emergency Management. Each emergency is unique, the level of weather support required for each situation will be dictated by appropriate 52 FW, Emergency Management (EM), and OSW leadership. The following section outlines specific duties and responsibilities related to 52 OSS/OSW EM support. The 52 OSS/OSW supports the 52 FW with the following crisis response actions:

2.5.1. Assists the Installation Commander and EM personnel in educating installation agencies on the purpose, applicability, and operating procedures of the watch and warning system and types of severe weather threats to the local area.

2.5.2. Provides severe weather guidance for developing the Installation Emergency Management Plan.

2.5.3. Provides weather data upon request to the Chemical, Biological, Radiological, Nuclear, and Explosives Control Center and to on-scene emergency personnel responding to a major accident. The 52 OSS/OSW will provide any and all-weather information requested for EM such as current wind speed and direction, temperature, relative humidity etc.

2.5.4. Develops Effective Downwind Messages and Chemical Downwind Messages (EDM/CDM). These are normally requested during FPCON Charlie or greater.

2.5.5. Performs formal reviews of severe weather events.

2.5.6. Provides weather observations and forecasts to support response operations.

2.5.7. Advises 52 FW/CC or representative on severe weather response actions and participates in annual preparedness meetings and exercises.

2.6. Aircraft Mishap. In the event of an aircraft mishap, the 52 OSS/OSW will:

2.6.1. Provide an immediate update to the Supervisor of Flying (SOF) and TOP 3.

2.6.2. Complete a local data save from 12 hours prior through 6 hours after the event. Reference DAFMAN 15-129, paragraph 5.7 for specific information required.

2.6.3. Initiate a data save through the BIFRÖST Portal for the time of the Mishap.

2.6.4. All data will be safeguarded locally and disseminated only to required agencies by weather flight leadership IAW DAFI 91-204, AFMAN 10-206, and local policies.

2.7. Air Traffic Control (ATC) Limited Observation Training. 52 OSS/OSW manages the ATC Limited Observation Training program. Weather personnel task-certify ATC personnel to evaluate prevailing visibility values from the control tower. If required, weather personnel will also ensure ATC personnel can operate the applicable weather equipment in ATC facilities. Log ATC task certification on ATC provided AF Form 3622, Air Traffic Control/Weather Certification and Rating Record. This training is designed to enable the Cooperative Weather Watch program and ensure "eyes forward" capabilities, IAW AFMAN 13-204v3 **Para. 11.5**.

2.8. Flight Information Publications (FLIPs) Update. The 52 OSS/OSW is responsible for ensuring all weather information in the FLIPs (e.g., operating hours and equipment limitations) is current and accurate. The 52 OSS/OSW will review new FLIPs as soon as possible for changes to weather information, airfield take-off, landing, radar instrument approach minima, NOTAMs, and other key items which may drive changes to weather support. The 52 OSS/OSW will incorporate any changes to the FLIPs into local procedures as soon as possible. As applicable, the weather flight chief or designee will route any updates regarding weather support to Airfield Management (52 OSS/OSAM) for processing. Additional guidance pertaining to FLIPs can be found in AFI 11-215, *Flight Manuals Program*.

2.9. Additional Staff Weather Support. The 52 OSS/OSW staff will also provide the following weather support upon request:

2.9.1. Instrument Refresher Course (IRC). 52 OSS/OSW will provide a forecaster to present the weather portion of the IRC. The 52 OSS/OSW will provide refresher training for assigned pilots on METAR code, TAF code, MEF products, space weather, and other topics as arranged by the 52 OSS/OSW and 52d Operations Group Standardization and Evaluation (52 OG/OGV) office.

2.9.2. Mobility Concept Briefing. The 52 OSS/OSW will provide pre-deployment planning weather information at Mobility Concept Briefings. If required, this briefing may be presented as a weather annex to a TASKORD or OPORD IAW JP 3-59. The briefing will include forecasted departure conditions, enroute weather, anticipated conditions for destination arrival time, deployed area climatological conditions, and other topics briefed as required.

2.9.3. Quarterly safety meetings and SOF Meetings. 52 OSS/OSW will provide seasonal weather briefings at quarterly safety and SOF meetings when requested by the 52 FW Safety Office (52 FW/SE) or 52 OG/OGV.

2.9.4. Extended Weather Outlook. The 52 OSS/OSW creates an Extended Weather Outlook for Spangdahlem AB. The Spangdahlem AB Extended Outlook is produced daily Monday through Friday, excluding family days and holidays. This product is distributed via Outlook no later than 0735L and contains text and graphics for a plain language weather forecast for general planning purposes. See **Attachment 4**.

2.9.5. Staff Briefings and Studies. 52 OSS/OSW provides climatology briefings, weather studies, or weather reviews when requested through the chain of command.

2.9.6. 52 OSS/OSW Flight Commander, Flight Chief, or other representative will attend all Airfield Operations Board Meetings.

2.9.7. Solar/Lunar Data. 52 OSS/OSW can provide detailed solar and lunar data. Data includes sunrise, sunset, moonrise, moonset, civil twilight, nautical twilight, lunar illumination, moon phases, solar and lunar angles, and millilux (measurement of illumination) for any location world-wide.

2.9.8. Space Weather. 52 OSS/OSW will provide space weather updates when solar events are forecast to degrade communications or operations.

2.9.9. Tropical Weather Updates. 52 OSS/OSW will provide briefings to wing leadership when a tropical system is expected to impact any 52 FW operation world-wide. The 52 OSS/OSW will work with the regional OWS, which serves as the primary liaison between the tropical cyclone forecast centers and Weather Flights. 52 OSS/OSW will use the Tropical Cyclone Tropical Assessment Product produced by the OWS or tropical cyclone forecasts issued by other designated tropical cyclone centers outside of the AOR (i.e. National Hurricane Center, Central Pacific Hurricane Center, or Joint Typhoon Warning Center). 52 OSS/OSW should not deviate from the official forecast position, track, movement, maximum wind speed, or intensity trend.

2.9.10. Volcanic Events. 52 OSS/OSW will provide briefings to wing leadership when volcanic ash could affect operations.

Chapter 3

AIRFIELD WEATHER SERVICE (AWS)

3.1. General. The AWS section provides support to the SOF, ATC, 603 Air Operations Center (AOC) forecast and resource protection mission, staff weather services, and PMSV contacts. AWS will publish (and amend as required) a valid Terminal Aerodrome Forecast (TAF) during open airfield hours. Additionally, AWS will supplement and/or back-up automated observations from the FBWOS as required.

3.2. Meteorological Watch (METWATCH). METWATCH is used to provide an organized approach for weather personnel to maintain situational awareness of both current and future meteorological situations. Changes in the status of weather elements result in notification from 52 OSS/OSW or 557 WW to base agencies.

3.3. Cooperative Weather Watch (CWW). A CWW is the process whereby ATC personnel, flying units, and Security Forces personnel report observed weather conditions to the 52 OSS/OSW. The 52 OSS/OSW will reevaluate weather conditions when information reported from a reliable source differs from the last disseminated observation. The 52 OSS/OSW may include the conditions in the next observation or as a new SPECI observation. See **Chapter 8**, Reciprocal Support, for more information.

3.4. Official Observation Points/Sites.

3.4.1. While disseminating automated observations, the official point of observation is the sensor suite at the end of the active runway.

3.4.2. When augmenting observations, the official point of observation is the fourth-floor rooftop of Building 47 (accessible from room 401).

3.4.3. While at the Alternate Operating Location (AOL), the active airfield sensor suite remains the official point of observation if the FBWOS & TDAU are operational, and automated observations are being disseminated. If the airfield sensors are inoperative while personnel are operating at the AOL, the official point of observation is on the southeast edge of the parking lot next to Building 73.

3.5. Types of Observations.

3.5.1. The 52 OSS/OSW provides observing services 24-hours a day using the FBWOS. The FBWOS, in conjunction with the SCA, will disseminate all observations automatically.

3.5.2. Types of Observation. Refer to AFMAN 15-124, *Meteorological Codes,* for help decoding weather observation and forecast products.

3.5.3. A SPECI is an unscheduled observation completed and transmitted when any of the special criteria listed below for automated (including when augmenting) and manual weather flights have been observed. SPECIs will contain all data elements found in a METAR plus additional remarks which elaborate on data in the body of the report. All SPECI reports will be prepared and transmitted as soon as possible after the relevant criteria are observed. Special criteria will be recorded and disseminated IAW AFMAN 15-111. All SPECI criteria can be found in the IDP.

3.6. Augmentation. Augmentation is the process of having position-qualified weather technicians manually add to or edit data in an observation generated by a properly sited FBWOS. The two augmentation processes used are supplementing and back-up.

3.6.1. Supplementing. Supplementing is a method of manually adding meteorological information to an automated observation that is beyond the capabilities of the FBWOS to detect and/or report. Weather personnel are required to have a view of the airfield complex when supplementing an FBWOS. Weather personnel will be prepared to supplement observations during controlled airfield hours when the weather conditions in **Table 3.1** are observed and/or forecast to occur within 1 hour. Weather personnel are required to be prepared to supplement observations whenever a tornado watch is valid, or warning has been issued for tornadic activity. Weather personnel are not required to supplement observations outside controlled airfield hours for **Table 3.2** criteria.

Table 3.1. Mandatory Supplementary Weather Conditions.

Tornado (+FC)
Funnel Cloud (FC)
Waterspout (+FC)
Hail (GR)
Volcanic Ash (VA)
Ice Pellets (PL)
Visibility (VIS) < ¼ mile (400 meters)
Snow Depth
Tower VIS (When surface prevailing VIS or tower VIS is < 4 miles (6000 meters) and the tower VIS differs from surface VIS by a reportable value)

Table 3.2. Mandatory Supplementary Weather Conditions During Wing Flying.

Ceilings form below 2000 feet.
Visibility decreases to less than 3 miles (5000 meters).
Thunderstorms (occurring and not reported or reported and not occurring)
Precipitation (type occurring is different than reported, precipitation is occurring and not reported, or precipitation is reported but not occurring) ***
***Forecasters will augment precipitation type during Wing Flying hours.

3.6.1.1. Augmentation Responsibilities. To perform augmentation duties, the weather technician must maintain situational awareness of current weather conditions and FBWOS observations. The 52 OSS/OSW has developed Operational Risk Management (ORM) based augmentation procedures based on Tables **4.1** and **4.2** which are documented in flight SOPs with clearly defined duty priorities that are available on the IDP.

3.6.2. Back-up of an FBWOS. Back-up is the method of manually providing meteorological data to an FBWOS observation when the primary automated method is not operational or unavailable due to sensor and/or communication failure. Except for some automated remarks, back-up provides the same reporting capability provided by a fully functioning FBWOS.

3.6.2.1. Wind and pressure values from any piece of back-up equipment (e.g., AN/TMQ-53, hand-held devices, or other MAJCOM-approved deployable meteorological equipment).

3.7. Pilot-to-Metro-Service (PMSV) and PIREPs. 52 OSS/OSW provides pertinent weather data to all aircraft operating within Spangdahlem AB airspace. The 52 OSS/OSW's callsign is "Spangdahlem Metro" and the assigned UHF frequency is 284.425 MHz.

3.7.1. The 52 OSS/OSW will solicit Pilot Reports (PIREPs) during each PMSV contact with aircrew. PIREPs enhance forecasting services and flight safety throughout the region for pilots or other flight operations.

3.7.2. Long term outages of 6 hours or more will be annotated via NOTAM until repaired.

3.7.3. Phone patch is available through the 52 CP at 258.7 MHz.

3.7.4. Ramstein AB provides back-up PMSV support during their duty hours published in the FLIP supplement for Europe utilizing the same frequency as Spangdahlem (284.425 MHz) and the callsign "Ramstein Metro."

3.7.5. Aircraft requiring PMSV support during non-duty hours can phone patch the to the standby forecaster TACC Weather via 1-800-AIRMOBL (1-800-247-6625).

3.7.6. 52 OSS/OSW is the primary agency for disseminating PIREPS at Spangdahlem AB. For additional information regarding PIREPs or PIREP codes, refer to AFMAN 15-124, *Meteorological Codes*.

3.8. Space Weather. AWS forecasters forward received suspected space weather impacts to the 2d Weather Squadron for analysis and to other agencies which may be affected by the event.

Chapter 4

MISSION WEATHER INTEGRATION (MWI)

4.1. General. This chapter identifies local weapons systems, the most common missions and operating areas, and weather sensitivities associated with the organization, weapons systems, missions, and aircrews. Also, the MWI provides Mission Execution Forecasts (MEFs), flight weather briefs (FWB), MISSIONWATCH, MEF verification, and planning data for the 52 FW flying units.

4.2. Supported Unit/Missions/Requirements. The MWI provides direct weather support and FWBs to 52 FW units. All MWI certified forecasters will review AFMAN 11-202v3 and AFMAN 11-202v3 USAFESUP SPANGDAHLEMABSUP to become familiar with weather impacts to flight operations. The MWI will tailor operating hours to accommodate 52 FW flying windows. In general, the MWI forecaster arrives 1.5 hours prior to the first briefing time and is available until the termination of wing flying.

4.2.1. Location. The MWI briefer is located at the 480 FS, Building 360, in the Southwest corner of the lobby. The MWI forecaster may also perform duties from Room 401, Building 47, once all 480 FS pilots have stepped to their aircraft.

4.2.2. Contact Information. While embedded with the 480 FS, the MWI briefer can be reached at DSN 314-452-2132. Reference paragraph 1.8. for contact information when the briefer is not with the 480 FS.

4.2.3. Flight Weather Brief Documentation and Retention. All FWBs will be documented, maintained, and retained IAW the AF Records Disposition Schedule and AFMAN 33-322. FWB minimum required information, regardless of format, is directed in DAFMAN 15-129, attachment 2.

4.3. Geographic Area of Responsibility. The 52 OSS/OSW provides mission-tailored weather support for flying training areas used by units assigned or attached, i.e. Theater Support Packages (TSPs), to the 52 FW. The most up to date maps of these areas can be found locally in several locations (i.e. – WF 4th floor map, 480 FS Scheduling Office map, and close view of TRA 205/305 at the Top 3 Desk).

4.4. 52 FW Flying Mission Weather Limitations. The 52 OSS/OSW will coordinate weather limitations that affect flying missions with the 480 FS/DO at least once per year. The current list is in the IDP. These limitations will then be used to create MEFs for local flyers.

4.4.1. USAF General Weather Limitations. The following tables provide general, non-airframe dependent weather limitations.

Table 4.1. USAF General Weather Limitations.

Weather Condition	Impact	Customer Action
Ceiling < 2,000ft and/or Visibility < 3sm	IFR Alternate required	Add fuel to allow divert

4.4.2. Pilot Weather Categories (PWC). PWCs are designed to reduce the exposure of pilots with limited experience to the risks inherent during periods of low ceiling and visibility. Most current PWC are located in the Spangdahlem IDP.

Table 4.2. Event Weather Limitations.

Event	Minimum
Formation Takeoff	Standing water, ice, slush or snow is on the runway. The crosswind or gust component exceeds 15 knots
Formation landing	The crosswind or gust component exceeds 15 knots. The runway is wet or reported wet. Ice, slush, or snow on the runway. 500 FT / 1.5 SM (or a flight member's weather category, whichever ishigher)
VFR Rejoin	Day – 1,500 FT / 2 SM (cig/vis) Night – 3,000 FT / 5 SM (cig/vis)

4.5. Spangdahlem AB Non-Flying Mission Weather Limitations. 52 OSS/OSW is staffed primarily for flying operations, airfield support, and Wing staff integration. However, other organizations on base have missions that directly support the aforementioned functions and may require weather support at various times. 52 OSS/OSW has documented support agreements in the IDP for resource protection supporting non-flying operations. Additionally, SABI's with actions that directly relate to weather (namely environmental Emergency Management) contain documented support provided to other organizations involved, mostly within Mission Support Group. Any additional requests for weather support can be submitted as a Support Assistance Request or MOA, which will be weighed against the priorities mentioned prior.

4.6. MEFs. The MEF is the primary tool used to accomplish day-to-day weather support for 52 FW flying operations. MEFs are tailored to individual customer requirements and are developed using a 2-phase process outlined in DAFMAN 15-129 and local procedures. During this process, the 52 OSS/OSW will tailor and integrate products created by strategic and theater weather centers, as well as information supplied by local units (e.g., flying schedule) and agencies. The result is a product designed to provide timely, accurate, and relevant weather support to customers. The MEF should be consistent with OWS and 557 WW issued products. However, during rapidly changing conditions, emergencies, or when conditions threaten resource protection, the 52 OSS/OSW will verbally update the Top 3, as well as amend the MEF to accurately reflect conditions. The MEFP SOP is located on the 52 OSS/OSW SharePoint.

4.7. MISSIONWATCH. MISSIONWATCH is the process by which 52 OSS/OSW monitors the weather for all local missions. Combined with METWATCH, MISSIONWATCH allows 52 OSS/OSW to monitor operations around Spangdahlem AB, including regional flying areas and routes, to issue amendments to any MEF.

4.7.1. During rapidly changing conditions, and the MEF is not representative, 52 OSS/OSW will amend it to ensure the forecast accurately reflects current or expected conditions and contact the SOF and Top 3 to relay mission impacts. MISSIONWATCH will include determining how weather will affect defined air and ground mission-limiting parameters as listed in the IDP.

4.8. Forecast Amendments. An amendment to the MEF flying package is required whenever one or more of the criteria listed in **Table 4.3** occurs for a designated forecast valid period.

Table 4.3. MEF Amendment Criteria.

Amendment Criteria

Ceiling (FT)	2000	700	300
Visibility (M)	5000	3200	1600
Icing	Unforecast, change of intensity or no longer occurring		
Turbulence	Unforecast or no longer occurring (MDT or greater)		
TSTMs	Unforecast or no longer occurring		
Mission Wx	Anytime observed or forecast Go/No Go conditions. change from a Go to a No Go (or No Go to a Go) for that mission type (i.e., CAS, CSAR etc.…)		
Space Wx	Unforecast severe impacts to GPS or UHF		
Alternate Wx	Unforecast change that crosses 1000ft/3200m		
Note: Cig and Vis changes are for going above or below the listed criteria. Amendments will also be made to the take-off/landing data section of the MEF when the ETAD TAF is amended for crossing a TAF threshold.			

4.9. Off-station Support. The 52 OSS/OSW is the primary source for tailored weather information for off-station support for 52 FW aircraft. When mission activities occur TDY or at a deployed location, 52 OSS/OSW leadership will determine the most effective means of ensuring the 52 FW units are receiving the appropriate level of support.

4.9.1. The 52 OSS/OSW will provide MEFs to Spangdahlem units/assets, i.e., 480 FS, transiting or conducting flying missions at other locations by deploying a forecaster with the unit, via reach-back, or by arranging support through other weather units or the OWS servicing the off-station location.

4.10. Additional Aircrew Briefing Support.

4.10.1. Transient & AMC Aircrew Services. During controlled airfield hours, transient aircrews may receive flight weather briefings from 52 OSS/OSW with proper coordination according to the standard duty priorities outlined in the latest rendition of the Spangdahlem IDP. AMC aircrews filed with the Tanker-Airlift Control Center (TACC) for weather support will be required to contact TACC for weather updates.

4.10.2. 175-1 Briefings. The 52 OSS/OSW prepare DD Form 175-1, *Flight Weather Briefing*, in compliance with DAFMAN 15-129, Attachment 2

4.10.3. Mass Briefings. MWI forecasters will provide in-person mass weather briefs to the 52 FW assigned units, i.e., 480 FS and TSPs, as able.

4.10.4. Not Utilized.

4.10.5. CORONET Briefings. 52 OSS/OSW supports host CORONET aircraft movements departing from Spangdahlem AB with planning and mission weather briefings.

4.10.6. Tactical Decision Aids (TDAs). The 52 OSS/OSW requires a description of the target scenario and 2 hours advance notice to produce Electro Optic (EO) TDAs. The accuracy of the product relies on the detail of the operational input the user supplies. The 52 OSS/OSW uses Target Acquisition Weapons Software to produce tailored target aids. TDA data may be provided by the 52 OSS/OSW in mass briefings and tailored to the day's given missions, upon request.

Chapter 5

WEATHER WATCHES, WARNINGS, AND ADVISORIES (WWAS)

5.1. General.

5.1.1. 52 OSS/OSW will issue WWAs when any of the specific criteria listed in the Spangdahlem IDP is occurring or expected to occur, as appropriate, within a 5 NM radius (unless otherwise specified) from the Spangdahlem AB runway. **Attachment 2** outlines the WWA's along with their desired lead times, definitions, and supported agencies that receive weather WWA notifications from 52 CP.

5.1.2. WWAs will specify the magnitude of the weather event as applicable. For example, the operational threshold for a damaging wind event is > 50 knots; the forecaster will issue a weather warning and specify the peak wind speed, such as peak gust 65 knots. The forecaster will also specify maximum hail size and rain/snow accumulations meeting or exceeding the WWA threshold.

5.1.3. WWA Numbering System. WWAs will automatically be assigned a number sequentially based on the month and the number of previous. For example, Weather Warning 06-003 is the third warning issued in the month of June, regardless of the warning's stated weather phenomena or method of dissemination.

5.2. WWA Definitions.

5.2.1. Weather Watch. A weather watch is a special notice issued by the weather flight to alert agencies to the potential for weather conditions of such intensity to pose a hazard to life or property. The weather watch can be thought of as a "heads up," at which time agencies need to consider implementing required protective actions should a subsequent weather warning be issued.

5.2.1.1. Upon receipt of a weather watch, operations may continue. However, personnel in high-risk areas should be prepared to implement corresponding required protective actions without delay if and when a subsequent weather warning is issued.

5.2.1.2. Watch Termination. Each watch is issued with a beginning and ending time. Once issued, the forecaster monitors weather conditions and either upgrades the watch to a warning, extends the watch, cancels the watch, or allows it to expire if evaluation shows the phenomena will not occur.

5.2.2. Weather Warning. A weather warning is a special notice issued to agencies to provide advance notification (sufficient time for protective actions) of expected weather conditions of such intensity to pose a hazard to life or property. It is important to remember that Watches alert to *potential*, while Warnings alert to what is expected to happen in the near future based on observed satellite, radar, and weather sensor data.

5.2.2.1. Upon receipt of a weather warning, required protective actions corresponding to the weather event will be immediately implemented to safeguard personnel and equipment.

5.2.2.2. Warning Initiation. Each warning is issued by the 52 OSS/OSW forecaster with a beginning and ending time. Once issued, the forecaster monitors weather conditions and ensures the warning adequately describes the timing and intensity of the weather threat.

5.2.2.3. Lightning warnings are observed, so they will be issued without an end time (indicated by UFN "Until Further Notice") when observed by the 52 OSS/OSW within 5 NM of the base.

5.2.2.4. Warning Termination. The OSW forecaster cancels the warning (or allows it to expire) if evaluation shows the conditions have ceased and will not occur again or will not occur at all.

5.2.3. Weather Advisory. A weather advisory is a special weather product to alert an organization to the occurrence of, or imminent occurrence of, weather conditions impacting operations. An advisory is a notification of weather criteria that is less intense than watch or warning criteria, but still has a specific impact to a specified organization or mission.

5.2.3.1. Upon receipt of a weather advisory, the responsible agencies will implement required procedures. All weather advisories are based on specific observed criteria and are initiated by the 52 OSS/OSW to notify agencies of currently occurring weather conditions hazardous to operations.

5.2.3.2. Weather advisories are terminated when the condition is no longer observed or when PIREPs indicate the condition no longer exists.

Chapter 6

DISSEMINATION OF WEATHER INFORMATION

6.1. General. The 52 OSS/OSW disseminates observations, TAFs, WWAs to ATC facilities and 52 CP via JET.

6.2. Primary Issuance. The 52 OSS/OSW will issue forecasted WWAs via the Integrated Weather Warning Coordination (IWWC) system within JET. IWWC is an automated phone system that will relay the conditions of each WWA using a recorded voice. The IWWC system tracks Spangdahlem AB agency receipt acknowledgement for each WWA message.

6.3. Access. Agencies can view this data using the 52 OSS/OSW Spangdahlem AB's JET guest portal website at: **https://owsjet28.sc.afcent.af.mil/portal/private/GuestSpangdahlem/Sensors**. CAC-enabled computers are required to view the JET Portal.

6.4. External Products. MEF and other mission weather briefings are disseminated via the base shared drive located at: **https://usaf.dps.mil/sites/52og/480FS/Section%20Continuity/Forms/AllItems.aspx?viewid=ea9aba53%2Dcdca%2D4bf4%2D9bc8%2Da78a8d867208&id=%2Fsites%2F52og%2F480FS%2FSection%20Continuity%2FOSW%20%28Weather%29**

6.5. AOL Weather Dissemination. If the 52 OSS/OSW must evacuate to the AOL the technicians will notify agencies in **Table 6.1**. Standard products will continue to be received through the SCA and the base shared drive as long as the systems remain operational. The on-shift forecaster will continue to perform all shift responsibilities from the AOL.

6.6. Back-up. In the event of a SCA and/or IWWC outage, observations and WWAs will be disseminated locally using AF Form 3806 to ATC and 52 CP facilities via phone, e-mail, radio, or any means available. Back-up contact numbers can be found in **Table 6.1**. These products will be disseminated long-line via internet by requesting a theater weather flight or the 28 OWS to disseminate via their SCA and/or JET system.

Table 6.1. Phone Dissemination.

Order of Dissemination	Phone
Tower	452-7218
WF	452-6749
Tower	452-6685
Command Post	452-6141
MOC	452-6887
Airfield Management	452-6633
AMC MOC	452-8941
21 OWS	485-2132
Exercise MOC	452-4826

Chapter 7

BACK-UP SUPPORT

7.1. General. There are numerous scenarios that could cause an interruption of service from the 52 OSS/OSW. This section briefly describes how weather services will be provided should any such events occur. The 52 OSS/OSW will utilize the Primary, Alternate, Contingency, and Emergency Communication Plan to ensure responsibilities and actions in their Installation Data Plan are maintained.

7.1.1. **Primary:** NIPR (bldg. 47 or AOL 360 (or in conjunction with the 480 FS),

7.1.2. **Alternate:** VPN/Wi-Fi

7.1.3. **Contingency:** Request assistance for weather situational awareness or to disseminate weather products from another unit. Potential options: 86 OSS / Servicing OWS / AOC / 31 OSS / 48 OSS / 100 OSS

7.1.4. **Emergency:** Organization outside of those listed above capable of supporting weather requirements during outage.

Chapter 8

RECIPROCAL SUPPORT

8.1. 52 FW Command Post (CP).

8.1.1. Disseminates weather WWAs notifications as listed in **Attachment 2** so that agencies may take the necessary actions to minimize damage.

8.1.2. Transmits OPREP reports when required as a result of severe weather occurrences.

8.2. 52 FW Safety (SE).

8.2.1. Notifies the 52 OSS/OSW of the following:

8.2.1.1. Investigations of aircraft mishaps of assigned or other military aircraft the 52 FW investigates.

8.2.1.2. Investigations of ground accidents involving weather or weather services.

8.2.2. Coordinates with 52 OSS/OSW on all messages containing references to weather.

8.2.3. 52 FW/SE will inform 52 OSS/OSW of requirements for seasonal weather briefings at least 3 duty days in advance.

8.3. 52 FW Public Affairs (PA).

8.3.1. **Upon request, takes photographs covering a 360-degree view of the airfield complex from both the building 47 and AOL observing points for 52 OSS/OSW's visibility binder.**

8.4. 52 FW Plans and Programs (XP).

8.4.1. Informs the 52 OSS/OSW of and provides access to plans and programs which require weather planning and inputs.

8.5. 52d Operations Group (52 OG).

8.5.1. 52d Operations Group Standardization and Evaluation (52 OG/OGV).

8.5.1.1. Coordinate requirements for SOF and IRC briefings at least 3 duty days in advance.

8.5.1.2. Assists 52 OSS/OSW with validating local airfield and range mission weather limitations at least once annually.

8.5.1.3. Ensure SOF duties include procedures to:

8.5.1.3.1. Inform 52 OSS/OSW of any declared alternate airfields.

8.5.1.3.2. Provide any received PIREPs within the local flying area to 52 OSS/OSW within 5 minutes of receipt.

8.5.1.3.3. Receive weather update from the AWS prior to assuming SOF duties for their shift.

8.5.2. 480th Fighter Squadron (480 FS).

8.5.2.1. Ensures flying schedules are available to 52 OSS/OSW

8.5.2.2. Encourage aircrews and TOP 3 to pass PIREPs to 52 OSS/OSW or to ATC/SOF. PIREPs that include cloud amounts, bases, and tops, as well as turbulence and icing encountered are particularly valuable.

8.5.2.3. Provide feedback/mission debrief to 52 OSS/OSW.

8.5.2.4. Coordinate weather support for all exercises, deployments, or other off station needs with 52 OSS/OSW as soon as possible.

8.5.2.5. Ensure the forecaster is incorporated into the planning phase of flying operations.

8.5.2.6. Report to 52 OSS/OSW any suspected space weather impacts to operations (for example, HF radio not functioning during period of solar activity).

8.5.2.7. Coordinate CORONET mission support at least 3 duty days in advance.

8.5.2.8. Coordinate Flight Weather Briefings, DD Form 175-1, at least 24 hours in advance.

8.5.2.9. Provide target details for the optimal development of TDAs.

8.5.3. 52 OSS Airfield Operations (52 OSS/OSA). This section outlines responsibilities of OSW and OSA as part of the Spangdahlem AB Cooperative Weather Watch (CWW) program. If duty conflicts arise with any requirements in the following sections, personnel will consult documented duty priorities and use ORM to complete tasks in order of highest priority.

8.5.3.1. Airfield Management (52 OSS/OSAM).

8.5.3.1.1. Notifies 52 OSS/OSW when updated Flight Information Publications (FLIPs) have been delivered.

8.5.3.1.2. Reports runway conditions (RCR) to 52 OSS/OSW.

8.5.3.1.3. Provides airfield management orientation for weather personnel.

8.5.3.2. Ground Controlled Approach (GCA) (52 OSS/OSAR).

8.5.3.2.1. Provides GCA orientation for weather personnel.

8.5.3.2.2. Relays all PIREPs to AWS technicians within five minutes of receipt for local and long-line dissemination IAW DAFMAN 15-129.

8.5.3.2.3. Reports SCA, JET and FBWOS malfunctions to 52 OSS/OSW.

8.5.3.2.4. Notifies 52 OSS/OSW when GCA personnel/facilities evacuate/relocate.

8.5.3.3. Air Traffic Control Tower (52 OSS/OSAT).

8.5.3.3.1. Provides tower orientation for weather personnel.

8.5.3.3.2. Takes tower visibility observations (tower VIS) when surface prevailing visibility or tower VIS is < 4 statute miles (6000 meters) and the tower VIS differs from surface visibility by a reportable value.

8.5.3.3.3. Relays all PIREPs and Tower Visibility observations to the OSW technicians within five minutes of receipt for local and long-line dissemination.

8.5.3.3.4. When requested, provides a Pilot-to-Metro Service (PMSV) radio check.

8.5.3.3.5. Relays weather WWAs to aircrews.

8.5.3.3.6. Relays obstructions to visibility.

8.5.3.3.7. Reports when lightning is observed.

8.5.3.3.8. Reports SCA, JET and FBWOS malfunctions to 52 OSS/OSW.

8.5.3.3.9. Notifies 52 OSS/OSW technicians of the following:

8.5.3.3.9.1. Runway lights become inoperative, or if inoperative, become operational.

8.5.3.3.9.2. Tower personnel/facilities evacuate/relocate.

8.5.3.3.9.3. Active runway changes.

8.5.3.3.9.4. Termination of wing flying.

8.5.3.4. Airfield Systems (52 OSS/OSACR).

8.5.3.4.1. Maintains the AN/FMQ-19, AN/TMQ-53, and PMSV radio.

8.5.3.4.1.1. RAWS will perform annual inspection of all meteorological equipment, ensure equipment is in good condition, and verify no obstructions are affecting the equipment siting and exposure. Weather leadership at the airfield will accompany airfield systems personnel on this inspection and log any new equipment limitation(s) into appropriate Flight Information Handbook or report equipment issues to higher level(s) (e.g., MAJCOM)

8.5.3.4.1.2. RAWS will certify the AN/FMQ-19 annually.

8.5.3.4.1.3. Notify 52 OSS/OSW at least 30 minutes prior to any preventative maintenance and on completion.

8.5.3.4.2. Provides a tour of airfield sensors for new weather personnel.

8.5.3.5. Airfield Automation (52 OSS/OSAX).

8.5.3.5.1. The Airfield Operations Automation Manager or AOAM / NCOIC, Airfield Automation Manager or NAAM Acts as a focal point for all non-JET and SCA related Airfield Automation System, also known as just AFAS, issues.

8.5.3.6. Weather Support to Airfield Operations.

8.5.3.6.1. 52 OSS/OSW will notify Tower and GCA of the following:

8.5.3.6.1.1. When weather personnel evacuate/relocate to the AOL.

8.5.3.6.1.2. Not Utilized.

8.5.3.6.1.3. Not Utilized.

8.5.3.6.1.4. When SCA, JET or FBWOS equipment malfunctions and when the equipment must be taken down for troubleshooting or maintenance.

8.5.3.6.2. Limited Observation Training. All new ATC personnel will call the 52 OSS/OSW and schedule an appointment with the 52 OSS/OSW Training Manager for local familiarization. At the end of training, ATC personnel are tested for qualification

and will receive a familiarization tour of the weather station. Orientation and certification will be conducted by certified weather personnel and documented on the AF Form 3622, ATC/Weather Certification and Rating Record. Training will cover the following:

8.5.3.6.2.1. Basic observing techniques.

8.5.3.6.2.2. Visibility and obstructions to visibility.

8.5.3.6.2.3. Local weather phenomena.

8.5.3.6.2.4. Decoding observations and TAFs.

8.5.3.6.2.5. SCA operations and FBWOS overview, including automated and augmented observation procedures.

8.5.3.6.2.6. Weather WWAs.

8.5.3.6.2.7. Importance of a cooperative OSW/OSA relationship.

8.5.3.6.3. Provide Tower facilities RVR readings when reported. If SCA is inoperative, OSW will update GCA and Tower of any changes to RVR until SCA and/or FBWOS have been restored.

8.5.3.6.4. Act as focal point to resolve all SCA and FBWOS outages. Pass pertinent weather data to GCA and Tower via telephone until SCA/FBWOS outages are resolved. In the event of telephone outage, time-sensitive weather data (i.e., RVR, rapidly changing ceiling, rapidly changing visibility, etc.) can be passed via PMSV while the SCA/FBWOS is being augmented.

8.5.3.6.5. Validate the Tower visibility charts at least annually and provide Tower with an MFR after the inspection to keep in their visibility binder IAW AFMAN 13-204v3 USAFE-AFAFRICASUPPLEMENT Para. 3.2.4.3.

8.5.3.6.6. Weather personnel will ensure supported ATC agencies are notified of all outages prior to contacting any maintenance agency.

8.5.4. 52d Operations Group Detachment 1, Łask AB, Poland (Det 1). Weather support for Det 1 is provided by the host nation, 28 OWS as defined in their IDP. This includes meteorological observations, TAFs, and WWAs. The 52 OSS/OSW will coordinate support when 52 FW jets are expected to operate out of Det 1.

8.5.4.1. Any changes to required weather support must be coordinated through the 52 OSS/OSW and documented on the IDP.

8.6. 52d Maintenance Group (52 MXG).

8.6.1. Maintenance Operations Center (MOC).

8.6.1.1. Receives weather WWAs from CP and notifies base maintenance agencies and Petroleum, Oil, and Lubricants Operations Facility.

8.6.1.2. Provides MOC orientation to newly assigned weather personnel.

8.6.1.3. Coordinates updates to weather portions of Emergency Actions (EA) Checklists with weather flight leadership and sends new copies of EA Checklists to weather flight leadership.

8.7. 52d Munitions Maintenance Group (52 MMG).

8.7.1. The 52 MMG has four Munitions Support Squadrons (MUNSS) which are geographically separated units (GSUs) located outside of Spangdahlem AB:

8.7.1.1. 701 MUNSS, Kleine Brogel AB, Belgium

8.7.1.2. 702 MUNSS, Buechel AB, Germany

8.7.1.3. 703 MUNSS, Volkel AB, Netherlands

8.7.1.4. 704 MUNSS, Ghedi AB, Italy

8.7.2. The 28 OWS is responsible for Resource Protection and maintains a separate IDP for each MUNSS to document required support.

8.7.3. Any changes to required weather support for any of the MUNSS should be coordinated through the 52 OSS/OSW and documented on the respective IDP.

8.8. 52d Mission Support Group (52 MSG).

8.8.1. 52d Communications Squadron.

8.8.1.1. Responsible for maintenance of phones, local area network, and internet communications.

8.8.1.2. 52 CS houses the JET/SCA Server but does not maintain server. Maintenance actions are performed by system admins within 52 OG.

8.8.1.3. 52 CS provides applicable TCNO patches required for the JET/SCA Server to system admins within 52 OG for maintenance to be performed.

8.8.1.3.1. Weather personnel will download the patches from the JET/SCA Support website and provide a link to their location on the shared network drives.

8.8.2. 52 Security Forces Squadron (52 SFS).

8.8.2.1. Military Working Dog Training. 52 OSS/OSW will provide weather information, to include observed and forecasted wind speed and direction, ambient temperature, and lightning data via the on-duty forecaster, as requested, to support 52 SFS working dog training.

8.8.2.2. Road Condition Status (ROADCON). 52 OSS/OSW will provide observed and forecasted snow, ice and heavy rain accumulation information as requested to aid the MSG/CC, MSG/CD, and 52 SFS in ROADCON determination. Note: 52 OSS/OSW does NOT make the ROADCON determination. It will only provide the necessary weather information required by appropriate authorities.

8.8.3. 52 Civil Engineer Squadron (52 CES).

8.8.3.1. Emergency Management (52 CES/CEX) prepares & maintains disaster preparedness plans for base survival and recovery during a natural disaster. 52 CES/CEX will coordinate any weather-related information regarding emergency plans with 52 OSS/OSW.

8.8.3.2. Snow and Ice Control/Removal (52 CES/CEOH) will notify 52 OSS/OSW to the scheduling of snow and ice control meeting, a pre-season and post-season snow brief.

8.8.3.3. In the event of a toxic chemical spill or release, 52 OSS/OSW will provide information, as requested, to 52 CES/CEX to be used by the Fire Department or CE Readiness Toxic Corridor programs.

8.8.3.4. In the event of an emergency, 52 CES/CEX will coordinate emerging WWA requirements as soon as possible.

8.8.3.5. The Fire Department will provide access for 52 OSS/OSW personnel to building 47.

8.8.3.6. 52 OSS/OSW will provide average temperatures for heating/cooling days to 52 CES, usually during the onset of fall and spring. Other specialized data is available upon request.

8.9. Tenant Units.

8.9.1. 726th Air Mobility Squadron (AMS). The 726 AMS is a tenant unit utilizing the Spangdahlem airfield and base facilities. The 618 AOC (TACC)/XOW and the OWS provide routine support to Air Mobility missions. Requests for weather support will be handled on a case-by-case basis IAW Attachment **2**, **52** OSS/OSW duty priorities.

8.9.2. Armed Forces Network Detachment 9 (AFN Det 9).

8.9.2.1. AFN broadcasts the current and forecast weather information over the radio throughout the day. This information can be found on the JET Guest Portal: **https://owsjet28.sc.afcent.af.mil/portal/private/GuestSpangdahlem/Sensors**.

8.9.2.2. 52 OSS/OSW will provide training for broadcasting basic weather information to AFN personnel upon request.

8.9.3. All 52 FW Agencies.

8.9.3.1. Notify 52 OSS/OSW of problems with JET equipment.

8.9.3.2. Notify 52 OSS/OSW, through proper chain of command, of new weather support requirements.

8.9.3.3. Each unit at Spangdahlem AB is responsible for coordinating additional WWA support or special notification for existing WWAs with 52 OSS/OSW. Each customer requesting support must validate the requirement by providing 52 OSS/OSW with a list of protective actions for each WWA.

8.9.3.4. Coordinate changes and additions to weather support requirements as soon as they are foreseen. Temporary non-routine requirements will be treated as Special Assistance Requests (SAR) and will be documented and supported to the best of 52 OSS/OSW capabilities. Depending on the duration and demand to resources, 52 OSS/OSW may request assistance from USAFE/A3AW or the 557 WW to aid with the needed support.

8.9.3.5. 52 OSS/OSW will provide the current IDP to all supported units upon request.

8.9.4. Transient Aircrew Support.

8.9.4.1. During controlled airfield hours, transient aircrews may receive flight weather briefings from 52 OSS/OSW according to the duty priorities outlined in the IDP.

8.9.4.1.1. The 52 OSS/OSW typically require a 24-hour advanced notice for flight weather briefings.

8.9.4.2. Flying units in which the 52 FW or 52 OG has operational control (OPCON) or tactical control (TACON), e.g. TSPs, should make every effort to provide or arrange for their own weather support prior to arriving in theater. This includes initiating a request for forces through their home MAJCOM.

8.9.4.2.1. While assigned to the 52 OG, flying units will follow the requirements listed for the 480 FS in section 8.5.2 if the 52 OSS/OSW is tasked to provide support.

NATHANIEL D. HOFMANN, Lieutenant Colonel
Commander

Attachment 1

GLOSSARY OF REFERENCES AND SUPPORTING INFORMATION

References

AFI 11-215, *Flight Manuals Program*, 25 Mar 2019

AFMAN 10-206, *Operational Reporting*, 18 Jun 2018, Incorporating Change 1, 01 Sep 2020

AFMAN 10-206 USAFE-AFAFRICA Supplement, *Operational Reporting*, 07 Sep 2023

AFMAN 11-202V3, *Flight Operations*, 10 Jan 2022

AFMAN 13-204V3, *Air Traffic Control*, 26 Apr 2024

AFMAN 13-204V3 USAFE-AFAFRICA Supplement, *Air Traffic Control*, 30 Mar 2023

AFMAN 15-111, *Surface Weather Observations*, 12 Mar 2019, Incorporating Change 1, 26 Jan 2024

AFMAN 15-124, *Meteorological Codes*, 16 Jan 2019

AFMAN 33-322, *Communication and Information*, 23 Mar 2020, Incorporating Change 1, 28 JUL 2021

DAFI 91-204, *Safety*, 10 Mar 2021

DAFMAN 15-129, *Air and Space Weather Operation*, 07 Sep 2023

DAFPD 15-1, *Weather Operations,* 28 May 2024

DoD Directive 5500.07, *Joint Ethics Regulation*, 15 May 2024

FCM-H1, *Surface Weather Observations and Reports*, Jul 2019

JP 3-59, *Meteorological and Oceanographic Operations*, 10 Jan 2018

Prescribed and Adopted Forms

AF Form 3622, *Air Traffic Control/Weather Certification and Rating Record*

AF Form 3803, *Surface Weather Observations (METAR/SPECI)*

AF Form 3806, *Weather Watch Advisory Log*

DAF Form 847, *Recommendation for Change of Publication*

DD Form 175-1, *Flight Weather Briefing*

Abbreviations and Acronyms

AFI—Air Force Instruction

AFMAN—Air Force Manual

AFW—Air Force Weather

AMC—Air Mobility Command

AOL—Alternate Operating Location

ATC—Air Traffic Control

CDM—Chemical Downwind Message

CP—Command Post

CWW—Cooperative Weather Watch

DAFMAN—Department of the Air Force Manual

DoD—Department of Defense

EA—Emergency Actions

EDM—Affective Downwind Message

EM—Emergency Management

EO—Electro Optics

EOC—Emergency Operations Center

FBWOS—Fixed Based Weather Observing System

FLIP—Flight Information Publication

FW—Fighter Wing

FWB—Flight Weather Brief

GSU—Geographically Separated Unit

IAW—In Accordance With

IWWC—Integrated Weather Warning Coordination

KTS—knots

LAN—Local Area Network

LOA—Letter of Agreement

JET—Joint Environmental Toolkit

MEF—Mission Execution Forecast

MEFP—Mission Execution Forecast Process

METAR—Aviation Routine Weather Report

METWATCH—Meteorological Watch

MOA—Memorandum of Agreement

MOC—Maintenance Operations Center

MWI—Mission Weather Integration

NM—Nautical Mile

NOTAM—Notice to Airmen

OPR—Office of Primary Responsibility

OPREP—Operational Report

ORM—Operational Risk Management

OSS/OSW—52d Operations Support Squadron Weather Flight

OWS—Operational Weather Squadron

PIREP—Pilot Report

PMSV—Pilot-to-Metro Service

RCR—Runway Condition Reading

RVR—Runway Visual Range

SABI—Spangdahlem Air Base Instruction

SCA—-Sensor Collection Appliance

SAR—Special Assistance Request

SOF—Supervisor of Flying

SOP—Standard Operating Procedure

SPECI—Aviation Special Weather Report

SWAP—Severe Weather Action Procedures

SWAT—Severe Weather Action Team

TACC—Tanker-Airlift Control Center

TAF—Terminal Aerodrome Forecast

TDA—Tactical Decision Aid

TSPs—Theater Support Packages

UHF—Ultra High Frequency

USAFE—United States Air Forces in Europe

WWAs—Watches, Warnings, and Advisories

Terms

557th Weather Wing (557 WW)—A strategic weather center at Offutt AFB NE, (formerly HQ Air Force Weather Agency (AFWA)) providing atmospheric data and analysis/forecast products required by the regional OWSs and the WFs worldwide. The 557 WW provides the centralized repository for global observations and forecasts that are data based at 557 WW and, in turn, disseminated to DOD weather data users worldwide. In addition to global observations and forecasts collected from worldwide sources, the 557 WW collects meteorological satellite data from multiple sources. Based on global analysis of available data, 557 WW creates global analysis and forecast products to meet the forecast requirements of its supported users.

Meteorological Watch—The monitoring of weather for a designated military operating area and informing supported agencies when certain weather conditions could affect operations.

Mission Execution Forecast—A MEF is the integration of strategic center products and perishable weather data to support an operator's weapon systems and tactics.

MISSIONWATCH—Monitoring the weather within a route, sortie, or training area and advising organizations when forecast or observed conditions breach operational thresholds or when hazardous weather conditions affect operations or pose a threat to life or property.

Operational Weather Squadron—An organization with regional forecast responsibility comprised of management, staff, and technical personnel and its assigned resources. Their mission is to produce fine-scale tailored weather forecast products and services to customers within their area of responsibility.

Weather Advisory—A special notice provided to a supported agency when an established weather condition is occurring that could affect its operation.

Weather Warning—A special notice provided to supported customers that alerts them of weather conditions (occurring or expected to occur within 5 NM of the center point of the runway) of such intensity as to pose a hazard to life or property.

Weather Watch—A special notice provided to supported customers that alerts them of conditions favorable for the development (potential) of weather conditions of such intensity as to pose a hazard to life or property.

Attachment 2

WWA NOTIFICATION MATRICES

Table A2.1. Watches, Warnings and Advisories.

Watch Criteria			
Lightning within 5NM	---	30 minutes	52 OSS/OSW
Tornadoes	---	As potential warrants	52 OSS/OSW
Severe Thunderstorm	---	4 hours	52 OSS/OSW**
Damaging Winds	---	4 hours	52 OSS/OSW**
Freezing Precipitation	---	As potential warrants	52 OSS/OSW
Snow	---	As potential warrants	52 OSS/OSW
Heavy Snow	---	As potential warrants	52 OSS/OSW***
Heavy Rain	---	As potential warrants	52 OSS/OSW
Warning Criteria			
Lightning within 5NM of the installation	Observed	Observed	52 OSS/OSW*
Tornadoes	Forecast	15 minutes	52 OSS/OSW
Severe Thunderstorm	Forecast	2 hours	52 OSS/OSW**
Moderate Thunderstorm	Forecast	2 hours	52 OSS/OSW**
Damaging Winds GTE 50 knots	Forecast	2 hours	52 OSS/OSW
Strong Winds GTE 35 knots	Forecast	90 minutes	52 OSS/OSW
Freezing Precipitation	Forecast	90 minutes	52 OSS/OSW
Heavy Snow	Forecast	90 minutes	52 OSS/OSW***
Heavy Rain	Forecast	90 minutes	52 OSS/OSW***
Blizzard	Forecast	60 minutes	52 OSS/OSW
Advisory Criteria			
Criteria	**Forecast/Observed**	**Desired Lead Time**	**Issued By**
Snow	Forecast	90 Minutes	52 OSS/OSW***
SFC winds GTE 25kts (AUTO)	Observed	Observed	52 OSS/OSW*
Crosswinds GTE 20 knots with wet runway (Aviation Only)	Observed	Observed	52 OSS/OSW
Crosswinds GTE 25 knots with dry runway (Aviation Only)	Observed	Observed	52 OSS/OSW
Low level wind shear exists (reported from PIREP) (Aviation Only)	Observed	Observed	52 OSS/OSW
Icing exists (any intensity) below 10,000 feet (reported from PIREP) (Aviation Only)	Observed	Observed	52 OSS/OSW
Fighter Index of Thermal Stress (FITS) (Caution, Danger, Cancel Zones) (Aviation Only)	Observed	Observed	52 OSS/OSW
Ice FOD conditions for F-16	Observed	Observed	52 OSS/OSW*
WF Evacuation to Alternate Operating Location (AOL)	Observed	Observed	52 OSS/OSW
Lightning within 10NM (for F-35 act)	Observed	Observed	52 OSS/OSW
Lightning within 25NM (for F-35 act)	Observed	Observed	52 OSS/OSW

Table A2.2. Watches, Warnings, and Advisories Definitions.

Lightning	Lightning within 5 nautical miles
Tornado	Tornado / Funnel cloud (detected by radar or visually observed) AND threatening warned location / aerodrome complex
Severe Thunderstorm	Damaging Wind ETAD: GTE 50 kts -AND/OR- Damaging Hail ETAD: GTE 3/4inch
Moderate Thunderstorm	High Wind GTE 35 knots and less than severe criteria associated with thunderstorms -AND/OR- Large Hail GTE ¼ inch and less than severe criteria at the warned location aerodrome complex
Damaging Winds	Surface winds not associated with thunderstorms meeting or exceeding damaging wind threshold. ETAD: 50 knots
Strong Winds	Surface Winds not associated with Thunderstorms GTE 35 knots (or locally determined threshold) and less than severe threshold
Freezing Precipitation	Threshold determined by local emergency management agencies Default parameter: Liquid precipitation of any type and intensity falls and produces glaze ice on exposed surfaces
Snow	Threshold GTE 0.5" but LT 2" of snow in 12 hours as determined by local Emergency management agencies. New snowfall of a measurable amount (as determined by emergency management agencies) during a specified time period
Heavy Snow	Threshold 2" of snow in 12 hours as determined by local Emergency management agencies. New snowfall of a measurable amount (as determined by emergency management agencies) during a specified time period
Heavy Rain	Threshold 2" of rain in 12 hours as determined by local Emergency management agencies. New rainfall of a measurable amount (as determined by emergency management agencies) during a specified time period
Ice FOD conditions	Ambient temperatures between 20°F (-7°C) and 45°F (7°C) with precipitation (rain, fog, sleet, or snow) or Dewpoint within 9°F (5°C) of ambient temperatures between 25°F (-4°C) and 45°F (7°C) or Ambient temperature below 45°F (7°C) with standing water or a mixture of water with ice or snow within the immediate proximity of the engine inlet. Reference: T.O. 1-F-16A-I
Blizzard	All of the following must occur: a) Surface visibility less than or equal to ¼ mile/400 meters b) Considerable falling and/or blowing snow c) Sustained wind speeds or gusts ≥ 30 knots d) Duration > 3 hours

A2.2. 52 Command Post Weather Notifications. The 52 CP will contact the agencies listed in **Tables A2.2** and **Table A2.3**, through the appropriate announcement IAW their Quick Reaction Checklists.

Table A2.3. CP Weather Advisory Notifications.

Agency / Advisory	Surface winds ≥ 25kts	Crosswinds ≥ 20kts (Wet Runway)	Crosswinds ≥ 25 kts (Dry runway)	Low Level Wind Shear (From PIREP)	FITS	Ice Fod (F-16)	Icing Below FL100 (From PIREP)
FW/CC	X						
FW/CV	X						
OG/CC	X	X	X	X	X	X	X
MXG/CC	X				X	X	
MMG/CC	X						
MSG/CC	X						
MDG/CC	X						
FW/CCC							

FW/DS	X						
SFS/BDOC	X						
FIRE DISPATCH	X						
MOCC	X	X	X	X	X	X	X
SNOW CONTROL							
VEHICLE DISPATCH	X						
CS/CFP	X						
MDG/AMBULANCE SERVICES	X						
AFN DET 9	X						
SPANG COMMISSARY							
FLIGHT SIM							
GOLF COURSE	X						
BILLETING	X						

Table A2.4. CP Weather Watch and Warning Notifications.

Agency / Watch or Warning	Surface Winds 35-49 Kts	Surface Winds ≥ 50 Kts	Lightning w/in 5 NM of Spangdahlem AB	Tornado	Moderate T-storms	Severe T-storms	Heavy Precip (≥ 2" rain or snow in 12 hrs	Blizzard	Freezing Precip
FW/CC	X	X	X	X	X	X	X	X	X
FW/CV	X	X	X	X	X	X	X	X	X
OG/CC	X	X	X	X	X	X	X	X	X
MXG/CC	X	X	X	X	X	X	X	X	X
MMG/CC	X	X	X	X	X	X	X	X	X
MSG/CC	X	X	X	X	X	X	X	X	X
MDG/CC	X	X	X	X	X	X	X	X	X
FW/CCC			X	X	X	X	X	X	X
FW/DS	X	X	X	X	X	X	X	X	X
SFS/BDOC	X	X	X	X	X	X	X	X	X
FIRE DISPATCH	X	X	X	X	X	X	X	X	X
MOCC	X	X	X	X	X	X	X	X	X
SNOW CONTROL							X (Snow Only)	X	X
VEHICLE DISPATCH	X	X	X	X	X	X	X	X	X
CS/CFP	X	X	X	X	X	X	X	X	X
MDG/AMBULANCESERVICES	X	X	X	X	X	X	X	X	X
AFN DET 9	X	X	X	X	X	X	X	X	X

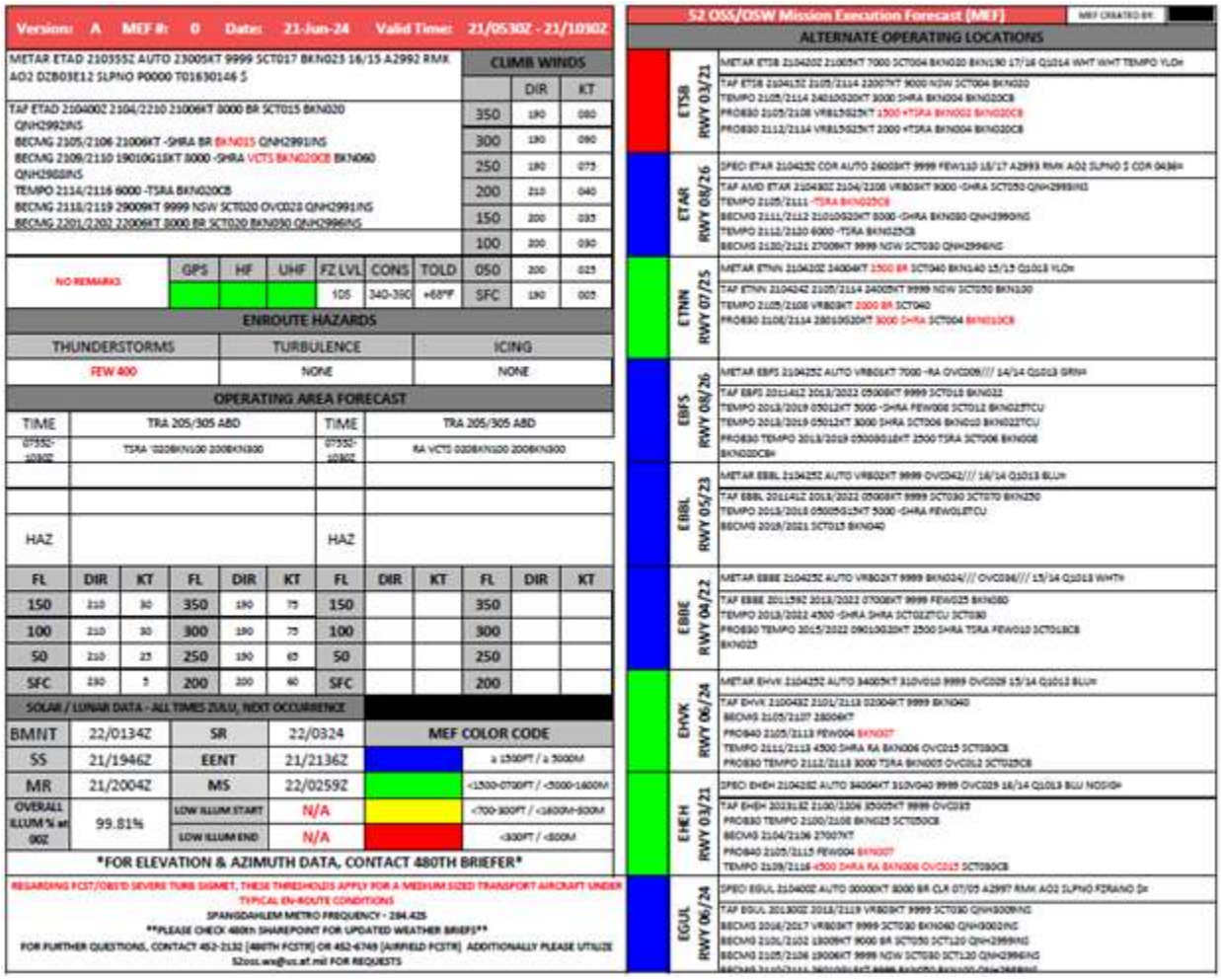
SPANG COMMISSARY			**X**	**X**	**X**	**X**	**X**	**X**	**X**
FLIGHT SIM			**X**	**X**	**X**	**X**	**X**	**X**	**X**
GOLF COURSE	**X**	**X**	**X**	**X**	**X**	**X**	**X**	**X**	**X**
BILLETING	**X**	**X**	**X**	**X**	**X**	**X**	**X**	**X**	**X**

A2.3.1. **Maintenance Operations Center (MOC) Weather Notifications** : MOC broadcasts all weather watches, warnings, and advisories over the maintenance nets, including the 52nd Logistics Readiness Squadron fuels flight, also known as Petroleum, Oil, and Lubricants.

Attachment 3

EXAMPLE MISSION EXECUTION FORECAST

Figure A3.1. Example Mission Execution Forecast.



Version: A MEF #: 0 Date: 21-Jun-24 Valid Time: 21/0530Z - 21/1030Z

METAR ETAD 210355Z AUTO 23005KT 9999 SCT017 BKN023 16/15 A2992 RMK AO2 DZB03E12 SLPNO P0000 T01630146 $

TAF ETAD 2104002 2104/2210 21006KT 8000 BR SCT015 BKN020
QNH2992INS
BECMG 2105/2106 21006KT -SHRA BR BKN015 QNH2991INS
BECMG 2109/2110 19010G18KT 8000 -SHRA VCTS BKN020CB BKN060
QNH2988INS
TEMPO 2114/2116 6000 -TSRA BKN020CB
BECMG 2118/2119 29006KT 9999 NSW SCT020 OVC028 QNH2991INS
BECMG 2201/2202 22006KT 8000 BR SCT020 BKN030 QNH2996INS

NO REMARKS

GPS	HF	UHF	FZ LVL	CONS	TOLD
			105	340-360	+68°F

CLIMB WINDS

	DIR	KT
350	190	080
300	190	090
250	190	075
200	210	040
150	200	035
100	200	030
050	200	025
SFC	190	005

ENROUTE HAZARDS

THUNDERSTORMS	TURBULENCE	ICING
FEW 400	NONE	NONE

OPERATING AREA FORECAST

TIME	TRA 205/305 ABD	TIME	TRA 205/305 ABD
0755Z-1030Z	TSRA 0208KN100 200BKN300	0755Z-1030Z	RA VCTS 0208KN100 200BKN300
HAZ		HAZ	

FL	DIR	KT	FL	DIR	KT	FL	DIR	KT	FL	DIR	KT
150	210	30	350	190	75	150			350		
100	210	30	300	190	75	100			300		
50	210	25	250	190	65	50			250		
SFC	190	5	200	200	60	SFC			200		

SOLAR / LUNAR DATA - ALL TIMES ZULU, NEXT OCCURRENCE

BMNT	22/0134Z	SR	22/0324	MEF COLOR CODE
SS	21/1946Z	EENT	21/2136Z	≥ 1500FT / ≥ 5000M
MR	21/2004Z	MS	22/0259Z	<1500-0700FT / <5000-1600M
OVERALL ILLUM % at 00Z	99.81%	LOW ILLUM START	N/A	<700-300FT / <1600M-800M
		LOW ILLUM END	N/A	<300FT / <800M

FOR ELEVATION & AZIMUTH DATA, CONTACT 480TH BRIEFER

REGARDING FCST/OBSTD SEVERE TURB SIGMET, THESE THRESHOLDS APPLY FOR A MEDIUM SIZED TRANSPORT AIRCRAFT UNDER TYPICAL EN-ROUTE CONDITIONS

SPANGDAHLEM METRO FREQUENCY - 284.425

PLEASE CHECK 480th SHAREPOINT FOR UPDATED WEATHER BRIEF!

FOR FURTHER QUESTIONS, CONTACT 452-2132 [480TH FCSTR] OR 452-6749 [AIRFIELD FCSTR] ADDITIONALLY PLEASE UTILIZE 52oss.ws@us.af.mil FOR REQUESTS

52 OSS/OSW Mission Execution Forecast (MEF) MEF CREATED BY:

ALTERNATE OPERATING LOCATIONS

ETSB RWY 03/21
METAR ETSB 2104202 21005KT 7000 SCT004 BKN020 BKN190 17/16 Q1014 WHT WHT TEMPO VLOI

TAF ETSB 2104152 2105/2114 12007KT 9000 NSW SCT004 BKN020
TEMPO 2105/2114 24010G20KT 3000 SHRA BKN004 BKN020CB
PROB30 2105/2108 VRB15G25KT 1500 +TSRA BKN002 BKN020CB
PROB30 2112/2114 VRB15G25KT 2000 +TSRA BKN004 BKN020CB

ETAR RWY 08/26
SPECI ETAR 2104252 COR AUTO 26003KT 9999 FEW110 18/17 A2993 RMK AO2 SLPNO $ COR 0436=

TAF AMD ETAR 2104302 2104/2108 VRB05KT 9000 -SHRA SCT050 QNH2985INS
TEMPO 2105/2111 -TSRA BKN025CB
BECMG 2111/2112 21010G20KT 8000 -SHRA BKN080 QNH2980INS
TEMPO 2112/2120 6000 -TSRA BKN025CB
BECMG 2120/2121 27006KT 9999 NSW SCT030 QNH2996INS

ETNN RWY 07/25
METAR ETNN 2104302 24004KT 1500 BR SCT040 BKN140 15/15 Q1018 VLOI

TAF ETNN 2104242 2105/2114 24005KT 9999 NSW SCT050 BKN100
TEMPO 2105/2108 VRB03KT 2000 BR SCT040
PROB30 2108/2114 28010G20KT 3000 SHRA SCT004 BKN010CB

EBFS RWY 08/26
METAR EBFS 2104252 AUTO VRB04KT 7000 -RA OVC006/// 14/14 Q1018 GRN=

TAF EBFS 2011412 2013/2022 05006KT 9999 SCT018 BKN022
TEMPO 2013/2019 05012KT 3000 -SHRA FEW006 SCT012 BKN022TCU
TEMPO 2013/2019 05012KT 3000 SHRA SCT006 BKN010 BKN022TCU
PROB30 TEMPO 2013/2019 05008G18KT 2500 TSRA SCT006 BKN008
BKN020CB=

EBBL RWY 05/23
METAR EBBL 2104252 AUTO VRB02KT 9999 OVC042/// 18/14 Q1018 BLU=

TAF EBBL 2011412 2013/2022 05006KT 9999 SCT030 3CT070 BKNZ50
TEMPO 2013/2018 05005G15KT 5000 -SHRA FEW018TCU
BECMG 2018/2021 SCT015 BKN040

EBBE RWY 04/22
METAR EBBE 2104252 AUTO VRB02KT 9999 BKN024/// OVC036/// 15/14 Q1018 WHT=

TAF EBBE 2011592 2013/2021 07006KT 9999 FEW025 BKN080
TEMPO 2013/2022 4500 -SHRA SHRA SCT022TCU SCT030
PROB30 TEMPO 2015/2022 09010G20KT 2500 SHRA TSRA FEW010 SCT018CB
BKN025

EHVK RWY 06/24
METAR EHVK 2104252 AUTO 34005KT 310V010 9999 OVC029 15/14 Q1013 BLU=

TAF EHVK 2100432 2101/2113 02004KT 9999 BKN040
BECMG 2105/2107 28004KT
PROB40 2105/2113 FEW004 BKN007
TEMPO 2111/2113 4500 SHRA RA BKN006 OVC015 SCT030CB
PROB30 TEMPO 2112/2118 3000 TSRA BKN005 OVC012 SCT025CB

EHEH RWY 03/21
SPECI EHEH 2104282 AUTO 34004KT 310V040 9999 OVC029 18/14 Q1013 BLU NOSIG=

TAF EHEH 2013182 2100/2106 35005KT 9999 OVC035
PROB30 TEMPO 2100/2108 BKN025 SCT050CB
BECMG 2104/2106 27007KT
PROB40 2105/2113 FEW004 BKN007
TEMPO 2109/2116 4500 SHRA RA BKN006 OVC015 SCT030CB

EGUL RWY 06/24
SPECI EGUL 2104002 AUTO 00000KT 8000 BR CLR 07/05 A2997 RMK AO2 SLPNO FZRANO $=

TAF EGUL 2013002 2018/2119 VRB03KT 9999 SCT030 QNH3009INS
BECMG 2018/2017 VRB03KT 9999 SCT030 BKN040 QNH3002INS
BECMG 2100/2102 18009KT 9000 BR SCT050 SCT120 QNH2989INS
BECMG 2105/2106 19006KT 9999 NSW SCT030 SCT120 QNH2996INS

Attachment 4

EXAMPLE EXTENDED FORECAST

Figure A4.1. Example Extended Forecast.

Friday, 21 Jun		Saturday, 22 Jun		Sunday, 23 Jun		Monday, 24 Jun		Tuesday, 25 Jun	
AM	PM	AM	PM	AM	PM	AM	PM	AM	PM
									
MOSTLY CLOUDY	CLOUDY	CLOUDY	MOSTLY CLOUDY	MOSTLY CLOUDY	MOSTLY CLOUDY	PARTLY CLOUDY	PARTLY CLOUDY	FEW	FEW
MIST & RAIN	T-STORMS	RAIN SHOWERS	RAIN	MIST & RAIN	RAIN	NONE	NONE	NONE	NONE
TRACE	0.10"	TRACE	TRACE	TRACE	0.10"				
5SM	4SM	6SM	6SM	2SM	6SM	7SM	7SM	7SM	7SM
1500FT	2000FT	2000FT	1700FT	700FT	2000FT				
61F / 16C	70F / 21C	52F / 11C	65F / 18C	52F / 11C	72F / 22C	56F / 13C	74F / 23C	59F / 15C	77F / 25C
									
SW	SW	SW	SW	NW	N	NE	NE	E	NE
06KT	10G18KT	10G15KT	10G20KT	04KT	04KT	06KT	06KT	06KT	09KT